

Bike Sharing Data Analysis – Summary Report

Project Overview

This project analyzes a bike-sharing system to understand usage patterns, station performance, and growth trends. The analysis uses cleaned data, proper data modeling, and Power BI visualizations to convert raw data into meaningful insights.

Project Objective

- To study bike usage across stations and cities
 - To identify high-performing and low-performing stations
 - To analyze trends over time
 - To support data-driven decision-making
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Data and Modeling

The dataset includes ride distance, station details, city information, and date attributes. Data was cleaned and organized into Fact and Dimension tables using a star schema. Relationships were created to ensure accurate analysis and reporting.

Data Preparation and Modeling

Before analysis, the data was:

- Cleaned to remove nulls and inconsistencies
- Standardized for naming and formats
- Split into Fact and Dimension tables
- Proper relationships were created using common keys

This structure improves performance, accuracy, and scalability in Power BI.

Descriptive Analysis (What happened?)

- Calculated total distance traveled and total revenue
- Identified stations and cities with highest and lowest usage
- Observed yearly and monthly usage trends

Insight: Bike usage is concentrated in a few key stations and cities.

Diagnostic Analysis (Why did it happen?)

- High usage is linked to high-demand locations
- Low performance is caused by limited demand or poor station placement
- Seasonal and yearly variations affect overall usage

Insight: Location and demand are the main drivers of performance.

Predictive Analysis (What may happen?)

- High-performing stations are expected to continue growing
- Cities with consistent growth will likely perform better in future
- Low-performing stations may remain underutilized

Insight: Future trends will likely follow historical patterns.

Prescriptive Analysis (What should be done?)

- Improve low-performing stations through better placement and promotion
- Focus investment on high-demand areas
- Monitor utilization and growth metrics regularly

Insight: Data-driven actions can improve efficiency and revenue.

Conclusion

This project demonstrates how data analysis and visualization can improve understanding of a bike-sharing system. By applying descriptive, diagnostic, predictive, and prescriptive analysis, the project provides clear insights that support better operational and strategic decisions.